

YEAR 7 MATHEMATICS

Chapter 1

Integers

WORKSHEETS

1

How to use this booklet

1 Each lesson is **one double-sided sheet** with six steps: **Easy** steps 1–2, **Medium** steps 3–4 and **Challenging** steps 5–6. Write your name at the top.

2 Each step starts with a **HOW TO** box and a worked example. The **speech bubble** tells you what to do.

WE DO Your teacher works these examples on the board. **Copy every step** onto the lines.

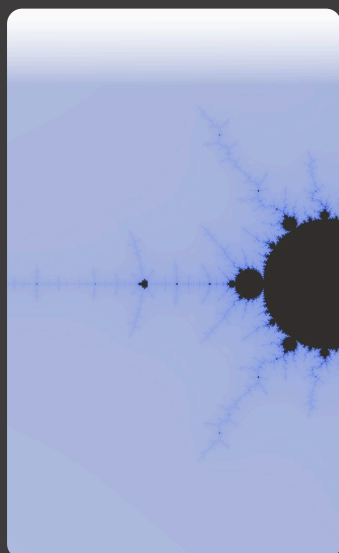
YOU DO Your turn. The first question is done for you, the next has the answer missing, then only the numbers are missing. Then it is all yours.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.01	Integers and opposites	3		☐
1.02	The number line	5		☐
1.03	Comparing and ordering integers	7		☐
1.04	Adding integers	9		☐
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In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



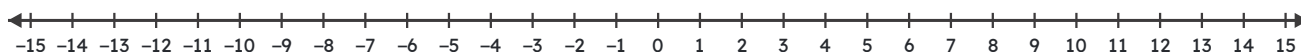
About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

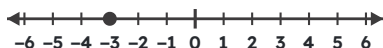


Step 1 What is an integer?

HOW TO EASY

An **integer** is a whole number. It can be positive, negative or zero. Negatives are left of 0. Positives are right of 0.

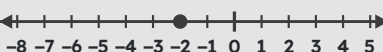
The dot is 3 left of 0. It is at **-3**.



Which integer is at the dot? Count from 0. The last one has no number line.

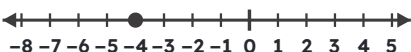
WE DO

E1 Where is the dot?



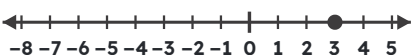
YOU DO

1 Where is the dot?



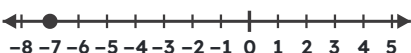
the dot is at **-4**

2 Where is the dot?



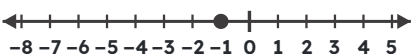
the dot is at ____

3 Where is the dot?



the dot is at ____

4 Where is the dot?



5 Which integer is 6 left of 0?

Step 2 Integers in real life

HOW TO EASY

Below zero, down, a loss or spending: **negative**.

Above zero, up, a gain or saving: **positive**.

5 m below sea level: **-5**

Write an integer for each one.

WE DO

E2 3°C below zero

YOU DO

6 a gain of 4 kg

7 8 m below sea level

= 4

= ____

8 a loss of \$12

9 6 floors down

= ____

10 a rise of 15 cm

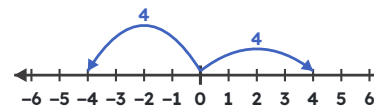
11 spending \$30

Step 3 Opposites

HOW TO MEDIUM

The **opposite** is the same distance from 0, on the other side.

The opposite of 4 is **-4**. The opposite of **-4** is 4.



Write the opposite of each integer.

WE DO

E3 7

12 -5

13 9

opposite: 5

opposite: ____

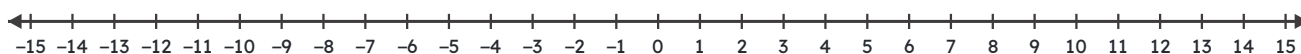
14 -12

15 30

opposite: ____

16 0

17 -100



Step 4 Double opposites

HOW TO MEDIUM

A $-$ in front of a bracket means the **opposite of**.

$-(-6)$ means the opposite of -6 , which is 6 .

$-(+6)$ means the opposite of 6 , which is -6 .

Say "the opposite of" for the $-$ in front of the bracket.



WE DO

E4 $-(-3)$

E5 $-(+4)$

YOU DO

18 $-(-8)$

19 $-(+5)$

= opposite of -8

= opposite of 5

= 8

= $---$

20 $-(-20)$

21 $-(+12)$

= opposite of $---$

= $---$

22 $-(-45)$

23 $-(+30)$

Step 5 Distance from zero

HOW TO CHALLENGING

The **distance from 0** is how many steps a number is from 0. It is never negative.

-7 or 5 : distances 7 and 5 .

Further: -7 .



Which is further from 0?
Write how far each one is.

WE DO

E6 -4 or 3

YOU DO

24 -8 or 5

distances: 8 and 5

further: -8

25 6 or -2

distances: 6 and 2

further: $---$

26 -10 or 9

distances: $---$ and $---$

further: $---$

27 -3 or -11

28 14 or -15

Step 6 Is it right?

HOW TO CHALLENGING

Work it out yourself first.

Then compare. Say yes or no, and give the right answer.

Kim says the opposite of -9 is 9 . Yes, Kim is right.

Is it right? Explain your answer.



WE DO

E7 Tom says $-(-4) = -4$. Is he right?

YOU DO

29 Ali says $-(+7) = 7$. Is he right?

= opposite of 7

= -7

No. The answer is -7 .

30 Ben says -6 is further from 0 than 4 . Is he right?

distances: 6 and 4

6 is more than 4

Yes / No. $---$

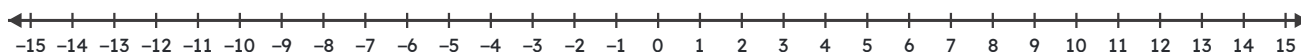
31 Mia says the opposite of 0 is 0 . Is she right?

$---$ is not left or right of $---$

Yes / No. $---$

32 Jo says 10 m down and 10 m up are opposites. Is Jo right?

NAME _____ CLASS _____ DATE _____



Step 1 Reading a number line

HOW TO EASY

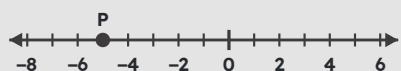
Numbers get larger to the **right**.
Only every second mark has a number. Count the marks from a number you know.

One mark left of -2 is -3.

Which integer is at the letter? The last one has no number line.

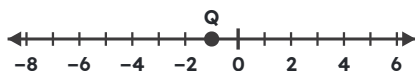
WE DO

E1 Where is P?



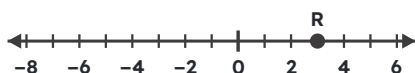
YOU DO

1 Where is Q?



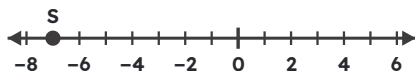
Q is at -1

2 Where is R?



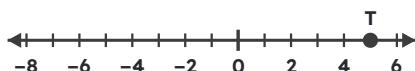
R is at ____

3 Where is S?



S is at ____

4 Where is T?



5 Which integer is 3 right of -5?

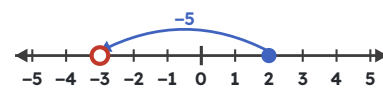
Step 2 Moving left and right

HOW TO EASY

Move **right**: the number gets larger.

Move **left**: the number gets smaller.

Start at 2, move 5 left. Land on -3.



Where do you land? Use the number line at the top of the page.

WE DO

E2 Start at 3. Move 5 left.

YOU DO

6 Start at -2. Move 6 right.

land on 4

7 Start at 1. Move 4 left.

land on ____

8 Start at -3. Move 5 left.

land on ____

9 Start at -7. Move 10 right.

10 Start at 6. Move 13 left.

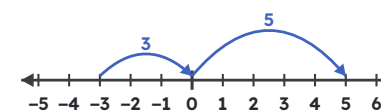
Step 3 Distance between

HOW TO MEDIUM

Count the steps from one number to the other.

Across 0? Count to 0, then on from 0.

-3 and 5: 3 to 0, 5 from 0. $3 + 5 = 8$.



How far apart are the two integers?

WE DO

E3 -2 and 4

E4 -7 and -3

YOU DO

11 -1 and 6

12 -5 and 3

1 to 0, 6 from 0

5 to 0, 3 from 0

= 7

= ____

13 -9 and -2

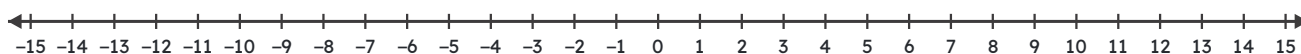
14 4 and -8

count ____ to ____

= ____

15 -12 and -5

16 -15 and 10



Step 4 Scales in 2s, 5s and 10s

HOW TO MEDIUM

Find the **step**: the difference between two labels ÷ the gaps between them.

-10 to 0 is 10, in 5 gaps. $10 \div 5 = 2$, step 2.

Find the step. Then find the number at the ?

WE DO

E5 What is at the ?



YOU DO

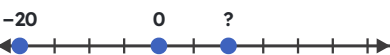
17 What is at the ?



$8 \div 4 = 2$, step 2

? = 4

18 What is at the ?



$20 \div 4 = 5$, step 5

? = 20

19 What is at the ?



$40 \div 5 = 8$, step 8

? = -12

20 The scale is -15, -10, □, 0. What is □?

Step 5 Halfway between

HOW TO CHALLENGING

Find how far apart they are. Halve it. Move that far from the smaller one.

-7 and 3: 10 apart, half is 5. $-7 + 5 = -2$.

Which integer is halfway between?

WE DO

E6 -4 and 2

YOU DO

21 -6 and 2

8 apart, half is 4

$-6 + 4 = -2$

22 -10 and 0

10 apart, half is 5

$___ + ___ = ___$

23 -3 and 9

$___$ apart, half is $___$

$___ + ___ = ___$

24 -11 and -1

25 -15 and 5

Step 6 Number line problems

HOW TO CHALLENGING

Picture a number line. Start there, then make each move.

An ant at -2 walks 5 left. It is at -7.

Where does it end up? Write each move.

WE DO

E7 An ant starts at 3. It walks 7 left. Where is it now?

YOU DO

26 A frog starts at -4. It hops 9 right. Where is it now?

start at -4, move 9 right

land on 5

27 Point A is at -6. Point B is 10 to the right of A. Where is B?

start at -6, move 10 right

land on 4

28 An ant at 5 walks 8 left, then 2 right. Where is it now?

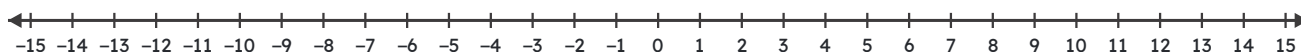
start at 5, move 8 left: -3

move 2 right

land on -1

29 A lift at level -3 goes up 7 floors, then down 6. Which level is it on?

NAME _____ CLASS _____ DATE _____

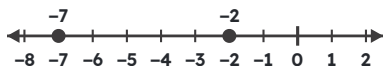


Step 1 Which is larger?

HOW TO EASY

On a number line, the number further **right** is larger.

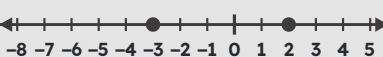
-2 is right of -7, so -2 is larger.



Which integer is larger?
The last one has no number line.

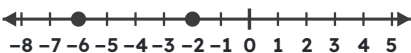
WE DO

E1 -3 or 2



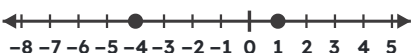
YOU DO

1 -6 or -2



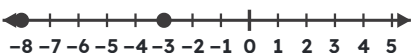
larger: **-2**

2 1 or -4



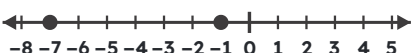
larger: **___**

3 -8 or -3



larger: **___**

4 -1 or -7



5 -12 or -9

Step 2 Using < and >

HOW TO EASY

< means **is less than**. > means **is greater than**.

The open side faces the larger number.

-5 < 2 and -1 > -6

Find the larger one. Then write < or > in the box.

WE DO

E2 -4 □ 3

YOU DO

6 -2 □ -6

7 0 □ -5

larger: **-2**

larger: **0**

symbol: **>**

symbol: **_____**

8 -9 □ -3

9 4 □ -10

larger: **___**

symbol: **_____**

10 -15 □ -14

11 -20 □ 1

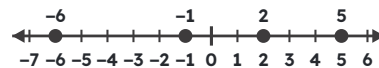
Step 3 Ordering

HOW TO MEDIUM

Ascending: smallest to largest, left to right on the number line.

Descending: largest to smallest.

Ascending: -6, -1, 2, 5



Find the smallest (or largest) first. Then write them in order.

WE DO

E3 Ascending: 3, -4, 0, -1

E4 Descending: -2, 5, -7, 1

YOU DO

12 Ascending: 2, -3, -1, 4

smallest: **-3**

-3, -1, 2, 4

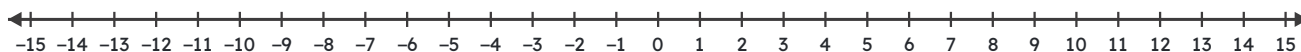
13 Descending: -4, 6, -1, 0

largest: **6**

14 Ascending: -12, 8, -15, -3, 1

smallest: **___**

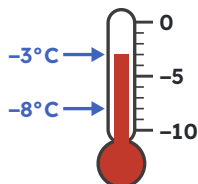
15 Descending: -20, -2, -11, 5, -8



Step 4 Comparing temperatures

HOW TO MEDIUM

Colder is **lower** on the thermometer: a smaller number.
 -8°C is colder than -3°C .



Which is colder (or warmer)? Picture the thermometer.

WE DO

E5 Which is colder: -4°C or -9°C ?

YOU DO

16 Which is colder: -2°C or -6°C ?

-6 is left of -2

colder: -6°C

17 Which is colder: 3°C or -1°C ?

-1 is left of 3

colder: -1°C

18 Which is warmer: -10°C or -7°C ?

-7 is right of -10

warmer: -7°C

19 Which is colder: -21°C or -12°C ?

Step 5 Simplify, then compare

HOW TO CHALLENGING

Simplify any $-(-n)$ or $-(+n)$ first.
 Then compare.
 $-(-2) \square -5$ becomes $2 \square -5$, so $>$.

Simplify. Then write $<$ or $>$.

WE DO

E6 $-(-3) \square -4$

YOU DO

20 $-(-1) \square -2$

21 $-(+4) \square -3$

$= 1 \square -2$

$= -4 \square -3$

symbol: $>$

symbol: $<$

22 $-(-5) \square 6$

23 $-7 \square -(+9)$

$= 5 \square 6$

symbol: $<$

24 $-(+6) \square 5$

25 $-8 \square -(-8)$

$= -6 \square 5$

$= -8 \square 8$

Step 6 Is it right?

HOW TO CHALLENGING

Work it out yourself first.
 Then compare. Say yes or no, and give the right answer.

Kim says $-3 > -8$. -3 is right of -8 . Yes, Kim is right.

Is it right? Explain your answer.

WE DO

E7 Tom says -9 is larger than -4 . Is he right?

YOU DO

26 Mia says $-1 > -10$. Is she right?

-1 is right of -10

so -1 is larger

Yes. Mia is right.

27 Ben says -20°C is colder than -2°C . Is he right?

-20 is left of -2

so -20°C is colder

Yes / No. Yes

28 Ali says $-5, -7, 0, 2$ is in ascending order. Is he right?

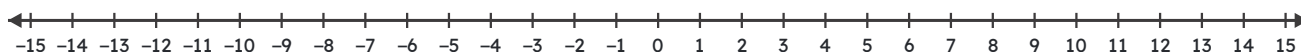
smallest: -7

$-7, -5, 0, 2$

Yes / No. It is $-7, -5, 0, 2$

29 Jo says $-(-6) < -6$. Is Jo right?

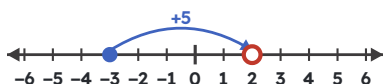
NAME _____ CLASS _____ DATE _____



Step 1 Adding a positive

HOW TO EASY

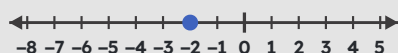
Start at the first number.
Adding a positive number: move **right**.
 $-3 + 5$: start at -3 , move 5 right.
Land on 2.



Draw the jump. Write where you land. The last one has no number line.

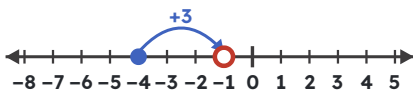
WE DO

E1 $-2 + 4$



YOU DO

1 $-4 + 3$



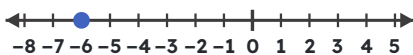
$= -1$

2 $-5 + 2$



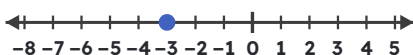
$=$ ____

3 $-6 + 4$



$=$ ____

4 $-3 + 7$

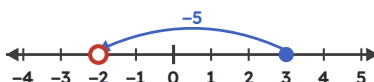


5 $-7 + 7$

Step 2 Adding a negative

HOW TO EASY

Adding a negative number: move **left**.
 $+(-4)$ means the same as -4 .
 $3 + (-5) = 3 - 5 = -2$



Rewrite $+(-)$ as $-$. Then work it out.

WE DO

E2 $6 + (-2)$

E3 $-2 + (-3)$

YOU DO

6 $7 + (-3)$

7 $5 + (-5)$

$= 7 - 3$

$= 5 - 5$

$= 4$

$=$ ____

8 $2 + (-6)$

9 $4 + (-10)$

$=$ ____

$=$ ____

$=$ ____

$=$ ____

10 $-1 + (-4)$

11 $-3 + (-6)$

$=$ ____

$=$ ____

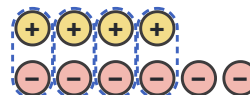
$=$ ____

$=$ ____

Step 3 Zero pairs

HOW TO MEDIUM

A $+$ and a $-$ make a **zero pair**.
They cancel out.
So take the difference. The sign with more wins.
 $-6 + 4$: $6 - 4 = 2$. $-$ has more: -2 .



Take the difference. Which sign has more?

WE DO

E4 $-5 + 8$

E5 $6 + (-9)$

YOU DO

12 $-7 + 10$

13 $-9 + 4$

$10 - 7 = 3$

$9 - 4 = 5$

$+$ has more: **3**

$-$ has more: ____

14 $8 + (-12)$

15 $-15 + 20$

$=$ ____

$=$ ____

$-$ has more: ____

16 $25 + (-40)$

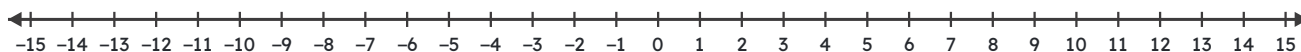
17 $-36 + 50$

$=$ ____

$=$ ____

$=$ ____

$=$ ____

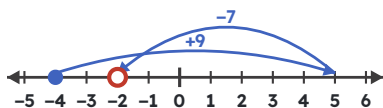


Step 4 Three numbers

HOW TO MEDIUM

Add from left to right, one jump at a time.

$$-4 + 9 + (-7) = 5 + (-7) = -2$$



Add from left to right.
Draw the jumps when
there is a number line.

WE DO

E6 $1 + (-4) + 5$



YOU DO

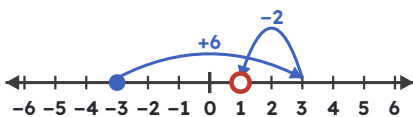
18 $2 + (-5) + 4$



$$= -3 + 4$$

$$= 1$$

19 $-3 + 6 + (-2)$



$$= 3 + (-2)$$

$$= \underline{\quad}$$

20 $12 + (-15) + (-6)$

$$= \underline{\quad} + (\underline{\quad})$$

$$= \underline{\quad}$$

21 $-20 + 8 + 15$

Step 5 Adding in real life

HOW TO CHALLENGING

Write the story as a sum.

Up, rises and gains are **positive**.

Down, falls and spending are **negative**.

-6°C , then it warms up 10°C :

$$-6 + 10 = 4^{\circ}\text{C}$$

Write a sum. Then answer
the question.

WE DO

E7 It is -5°C . The
temperature rises 8°C .
What is the temperature
now?

YOU DO

22 It is -4°C at dawn. By lunch
it has warmed up 9°C . What
is the temperature at lunch?

$$-4 + 9$$

$$= 5^{\circ}\text{C}$$

23 A lift is at level -2 . It goes
up 5 floors. Which level is it
on now?

$$-2 + 5$$

$$= \text{level } \underline{\quad}$$

24 Sam has $\$15$ in the bank.
Sam spends $\$40$ using
credit. What is the balance
now?

$$\underline{\quad} + (\underline{\quad})$$

$$= \underline{\quad}$$

25 A diver is at -18 m. She
swims up 7 m, then down
4 m. Where is she now?

Step 6 Is it right?

HOW TO CHALLENGING

Work it out yourself first.

Then compare. Say yes or no, and
give the right answer.

Kim says $3 + (-5) = -2$.

$3 - 5 = -2$. Yes, Kim is right.

Is it right? Explain your
answer.

WE DO

E8 Tom says $-6 + 2 = -8$. Is he
right?

YOU DO

26 Mia says $-8 + 5 = -13$. Is she
right?

start at -8 , move 5 right

$$= -3$$

No. The answer is -3 .

27 Ben says $-7 + 3 + (-3) = -7$.
Is he right?

$3 + (-3)$ is a zero pair

$$-7 + 0 = -7$$

Yes / No. $\underline{\quad}$

28 Ali says $-4 + \square = 2$ means $\square$
 $= -2$. Is he right?

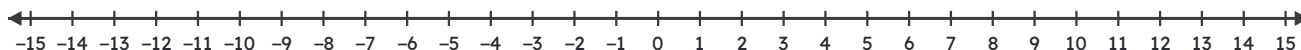
from $\underline{\quad}$ to $\underline{\quad}$ is $\underline{\quad}$ right

$$\square = \underline{\quad}$$

Yes / No. $\square$ is $\underline{\quad}$, not $\underline{\quad}$.

29 Jo says adding always
makes a number bigger. Is
Jo right?

NAME _____ CLASS _____ DATE _____



Step 1 Subtracting a positive

HOW TO EASY

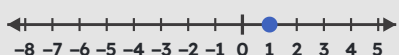
Start at the first number.
Subtracting a positive number:
move **left**.
2 - 5: start at 2, move 5 left.
Land on -3.



Draw the jump. Write where you land. The last one has no number line.

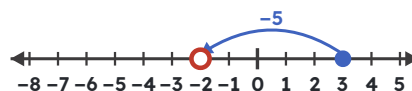
WE DO

E1 1 - 4



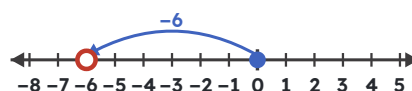
YOU DO

1 3 - 5



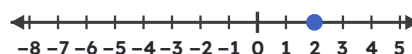
= -2

2 0 - 6



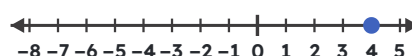
= ____

3 2 - 7



= ____

4 4 - 12



5 5 - 11

Step 2 Starting below zero

HOW TO EASY

Start below zero and subtract:
you go **further** below zero.
-3 - 4 = -7. You owe \$3, then
spend \$4 more: you owe \$7.

Move left. Write where you land.

WE DO

E2 -2 - 3

YOU DO

6 -1 - 4

move 4 left

= -5

7 -5 - 2

move 2 left

= ____

8 -6 - 6

move ____ left

= ____

9 -10 - 5

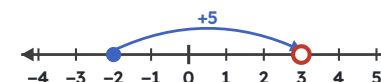
10 -8 - 12

11 -25 - 15

Step 3 Subtracting a negative

HOW TO MEDIUM

Subtracting a negative: move **right**.
- (-3) means the same as + 3.
-2 - (-5) = -2 + 5 = 3



Rewrite - (-) as +. Then work it out.

WE DO

E3 4 - (-2)

YOU DO

12 6 - (-3)

= 6 + 3

= 9

13 1 - (-7)

= 1 + 7

= ____

14 -2 - (-5)

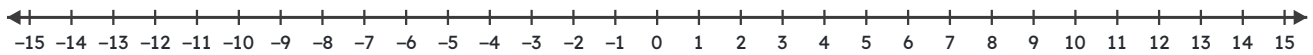
= ____ + ____

= ____

15 -8 - (-3)

16 -4 - (-4)

17 -15 - (-20)

**Step 4 Mixed + and -****HOW TO** **MEDIUM**

First rewrite each $- (-)$ as $+$.
Then work from left to right.

$$-3 - (-5) - 4 = -3 + 5 - 4 = 2 - 4 = -2$$

Rewrite $- (-)$ as $+$. Then work from left to right.

WE DO

E4 $2 - (-3) - 6$

YOU DO

18 $1 - (-4) - 7$

$$= 1 + 4 - 7$$

$$= 5 - 7$$

$$= -2$$

19 $-3 - 5 - (-2)$

$$= -3 - 5 + 2$$

$$= -8 + 2$$

$$= \underline{\hspace{2cm}}$$

20 $-6 - (-9) + (-5)$

$$= \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + (\underline{\hspace{1cm}})$$

$$= \underline{\hspace{1cm}} + (\underline{\hspace{1cm}})$$

$$= \underline{\hspace{2cm}}$$

21 $10 - (-2) - 15$

Step 5 Change and difference**HOW TO** **CHALLENGING**

Change = new - old. Difference = higher - lower.

From -4°C to 6°C : $6 - (-4) = 6 + 4 = 10$. A rise of 10°C .

Write a subtraction. Then answer the question.

WE DO

E5 It was -3°C , then 5°C .
How much did it rise?

YOU DO

22 It was -6°C , then 2°C . How much did it rise?

$$2 - (-6)$$

$$= 2 + 6$$

$$= 8^{\circ}\text{C}$$

23 A gull is at 25 m. A diver is at -15 m. How far apart are they?

$$25 - (-15)$$

$$= 25 + 15$$

$$= \underline{\hspace{2cm}} \text{ m}$$

24 A freezer is -18°C . The kitchen is 22°C . What is the difference?

$$\underline{\hspace{1cm}} - (\underline{\hspace{1cm}})$$

$$= \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$$

$$= \underline{\hspace{2cm}}^{\circ}\text{C}$$

25 It was 4°C , then -9°C . What was the change?

Step 6 Is it right?**HOW TO** **CHALLENGING**

Work it out yourself first.

Then compare. Say yes or no, and give the right answer.

Kim says $3 - (-2) = 5$. $3 + 2 = 5$.
Yes, Kim is right.

Is it right? Explain your answer.

WE DO

E6 Tom says $-4 - 3 = -1$. Is he right?

YOU DO

26 Mia says $2 - (-6) = -4$. Is she right?

$$= 2 + 6$$

$$= 8$$

No. The answer is 8.

27 Ben says $-5 - (-5) = 0$. Is he right?

$$= -5 + 5$$

$$= 0$$

Yes / No.

28 Ali says $-1 - 8 = -9$. Is he right?

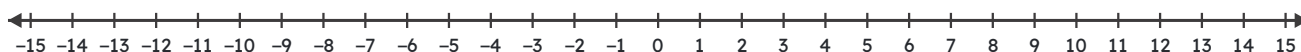
start at , move left

$$= \underline{\hspace{2cm}}$$

Yes / No.

29 Jo says subtracting always makes a number smaller. Is Jo right?

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Step 1 Equal groups

HOW TO EASY

$3 \times (-2)$ means 3 groups of -2 .
 $(-2) + (-2) + (-2) = -6$, so $3 \times (-2) = -6$.



Add the groups. Then write the answer.

WE DO

E1 $2 \times (-4)$

YOU DO

1 $2 \times (-3)$

$(-3) + (-3)$

$= -6$

2 $3 \times (-4)$

$(-4) + (-4) + (-4)$

$=$ ____

3 $4 \times (-2)$

$(\text{---}) + (\text{---}) + (\text{---}) + (\text{---})$

$=$ ____

4 $3 \times (-10)$

5 $2 \times (-25)$

Step 2 The sign rules

HOW TO EASY

Same signs: positive. **Different signs:** negative.

Multiply the numbers, then choose the sign.

-4×5 : **different signs:** $-$. $4 \times 5 = 20$, so -20 .

Choose the sign. Then multiply.

WE DO

E2 -3×4

YOU DO

6 $5 \times (-2)$

different signs: $-$

$= -10$

7 -6×3

different signs: $-$

$=$ ____

8 $-4 \times (-2)$

same signs: ____

$=$ ____

9 $7 \times (-5)$

10 $-9 \times (-3)$

11 -8×6

Step 3 Why $- \times -$ is $+$

HOW TO MEDIUM

Down the table, each answer goes up 3.

So a negative $\times$ a negative is positive.

product	answer
$1 \times (-3)$	-3
$0 \times (-3)$	0
$-1 \times (-3)$	3
$-2 \times (-3)$	6

Continue the pattern for two more lines.

WE DO

E3 $1 \times (-5) = -5$, $0 \times (-5) = 0$, ...

YOU DO

12 $1 \times (-4) = -4$, $0 \times (-4) = 0$, ...

$-1 \times (-4) = 4$

$-2 \times (-4) = 8$

13 $1 \times (-2) = -2$, $0 \times (-2) = 0$, ...

$-1 \times (-2) = 2$

$\text{---} \times (\text{---}) = \text{---}$

Now use the rule.

WE DO

E4 $-2 \times (-6)$

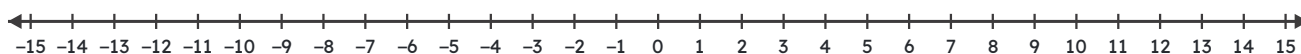
YOU DO

14 $-5 \times (-4)$

15 $-7 \times (-8)$

same signs: ____

$=$ ____

**Step 4 Three numbers****HOW TO****MEDIUM**

Multiply two numbers at a time, left to right.

Check: an **even** number of negatives gives +, an **odd** number gives -.

$$-2 \times 3 \times (-4) = -6 \times (-4) = 24$$

Multiply from left to right.

WE DO

E5 $2 \times (-3) \times 4$

YOU DO

16 $-2 \times 5 \times 3$

$$= -10 \times 3$$

$$= -30$$

17 $3 \times (-2) \times (-5)$

$$= -6 \times (-5)$$

$$= \underline{\hspace{2cm}}$$

18 $-3 \times 4 \times (-5)$

$$= \underline{\hspace{2cm}} \times (\underline{\hspace{2cm}})$$

$$= \underline{\hspace{2cm}}$$

19 $-4 \times (-2) \times (-3)$

20 $-5 \times (-2) \times (-2) \times (-3)$

Step 5 Multiplying in real life**HOW TO****CHALLENGING**

A change that repeats is a multiplication. Down, falling or spending is negative.

Down 3 m a minute for 5 minutes: $5 \times (-3) = -15$ m.

Write a multiplication. Then answer the question.

WE DO

E6 The temperature drops 2°C every hour for 4 hours. What is the total change?

YOU DO

21 A diver goes down 4 m every minute for 6 minutes. What is her change in height?

$$6 \times (-4)$$

$$= -24 \text{ m}$$

22 Sam spends \$5 a day for 7 days. What is the change in Sam's money?

$$7 \times (-5)$$

$$= \underline{\hspace{2cm}}$$

23 A share price falls \$3 a week for 9 weeks. What is the total change?

$$\underline{\hspace{2cm}} \times (\underline{\hspace{2cm}})$$

$$= \underline{\hspace{2cm}}$$

24 A lift at level 3 goes down 2 floors, 5 times. Which level is it on now?

Step 6 Is it right?**HOW TO****CHALLENGING**

Work it out yourself first.

Then compare. Say yes or no, and give the right answer.

Kim says $-4 \times (-5) = 20$. Same signs: +. Yes, Kim is right.

Is it right? Explain your answer.

WE DO

E7 Tom says $-3 \times 6 = 18$. Is he right?

YOU DO

25 Mia says $-7 \times (-2) = -14$. Is she right?

same signs: +

$$= 14$$

No. The answer is 14.

26 Ben says $-4 \times \square = 28$ means $\square = -7$. Is he right?

$$-4 \times (-7)$$

$$= 28$$

Yes / No.

27 Ali says $(-5)^2 = -25$. Is he right?

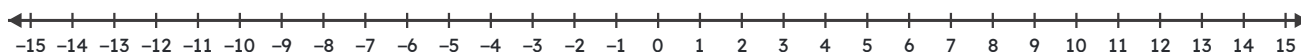
$$= \underline{\hspace{2cm}} \times (\underline{\hspace{2cm}})$$

$$= \underline{\hspace{2cm}}$$

Yes / No. The answer is .

28 Jo says $-2 \times (-3) \times (-4)$ is positive. Is Jo right?

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Step 1 Division undoes ×

HOW TO EASY

Every $\times$ fact gives two $\div$ facts.

$$3 \times (-4) = -12, \text{ so}$$

$$-12 \div 3 = -4 \text{ and } -12 \div (-4) = 3.$$

Write two $\div$ facts from each $\times$ fact.

WE DO

E1 $4 \times (-2) = -8$

YOU DO

1 $2 \times (-5) = -10$

$$-10 \div 2 = -5$$

$$-10 \div (-5) = 2$$

2 $3 \times (-6) = -18$

$$-18 \div 3 = -6$$

$$___ \div (___) = ___$$

3 $5 \times (-4) = -20$

$$___ \div ___ = ___$$

$$___ \div (___) = ___$$

4 $6 \times (-7) = -42$

Step 2 The sign rules

HOW TO EASY

The same as for $\times$. **Same** signs: positive. **Different** signs: negative.

$$-20 \div 4: \text{different signs: } -. \quad 20 \div 4 = 5, \text{ so } -5.$$

Choose the sign. Then divide.

WE DO

E2 $-18 \div 3$

YOU DO

5 $-12 \div 4$

different signs: -

$$= -3$$

6 $20 \div (-5)$

different signs: -

$$= ___$$

7 $-30 \div (-6)$

same signs: _____

$$= ___$$

8 $-42 \div 7$

9 $-56 \div (-8)$

10 $72 \div (-9)$

Step 3 Missing numbers

HOW TO MEDIUM

Use the opposite operation.

$$\square \div (-3) = 5, \text{ so } \square = 5 \times (-3) = -15.$$

$$-24 \div \square = 6, \text{ so } \square = -24 \div 6 = -4.$$

Find the missing number. Use the opposite operation.

WE DO

E3 $\square \div 2 = -4$

E4 $-20 \div \square = 5$

YOU DO

11 $\square \div 3 = -6$

12 $\square \div (-4) = 5$

$$\square = -6 \times 3$$

$$\square = 5 \times (-4)$$

$$\square = -18$$

$$\square = ___$$

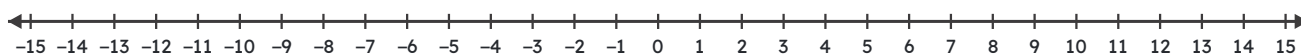
13 $-30 \div \square = -6$

14 $\square \div (-7) = -3$

$$\square = ___ \div (___)$$

$$\square = ___$$

15 $48 \div \square = -8$



Step 4 Dividing step by step

HOW TO MEDIUM

Divide from left to right, one step at a time.

$$-60 \div 5 \div (-3) = -12 \div (-3) = 4$$

Divide from left to right.

WE DO

E5 $-24 \div 2 \div 3$

YOU DO

16 $-40 \div 4 \div 2$

$$= -10 \div 2$$

$$= -5$$

17 $36 \div (-3) \div 2$

$$= -12 \div 2$$

$$= \underline{\hspace{2cm}}$$

18 $-48 \div (-4) \div (-3)$

$$= \underline{\hspace{2cm}} \div (\underline{\hspace{2cm}})$$

$$= \underline{\hspace{2cm}}$$

19 $-90 \div 3 \div (-5)$

Step 5 Sharing equally

HOW TO CHALLENGING

Sharing a debt or a fall equally: divide a negative.

A fall of 12°C over 4 hours: $-12 \div 4 = -3^{\circ}\text{C}$ each hour.

Write a division. Then answer the question.

WE DO

E6 A debt of \$20 is shared by 4 people. Write each share as an integer.

YOU DO

20 A debt of \$36 is shared by 3 people. Write each share as an integer.

$$-36 \div 3$$

$$= -\$12 \text{ each}$$

21 The temperature fell 15°C in 5 hours. What was the change each hour?

$$-15 \div 5$$

$$= \underline{\hspace{2cm}}^{\circ}\text{C each hour}$$

22 A diver went down 24 m in 6 minutes. What was her change each minute?

$$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}} \text{ m each minute}$$

23 A share price fell \$56 over 8 days, the same each day. What was the change each day?

Step 6 Is it right?

HOW TO CHALLENGING

Work it out yourself first.

Then compare. Say yes or no, and give the right answer.

Kim says $-18 \div (-3) = 6$. Same signs: +. Yes, Kim is right.

Is it right? Explain your answer.

WE DO

E7 Tom says $-24 \div 6 = 4$. Is he right?

YOU DO

24 Mia says $35 \div (-7) = -5$. Is she right?

different signs: -

$$= -5$$

Yes. Mia is right.

25 Ben says $-40 \div (-8) = -5$. Is he right?

same signs: +

$$= 5$$

Yes / No. The answer is ____.

26 Ali says $\square \div (-2) = 6$ means $\square = -3$. Is he right?

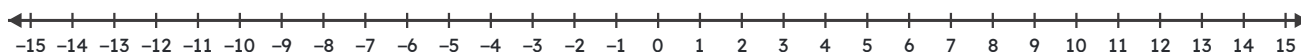
$$\square = \underline{\hspace{2cm}} \times (\underline{\hspace{2cm}})$$

$$\square = \underline{\hspace{2cm}}$$

Yes / No. $\square$ is ____, not ____.

27 Jo says dividing by a negative always gives a negative. Is Jo right?

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Step 1 The order

HOW TO EASY

1. Brackets. 2. $\times$ and $\div$, left to right. 3. $+$ and $-$, left to right.
 $-3 + 4 \times 2 = -3 + 8 = 5$

Do the $\times$ or $\div$ first. Then the $+$ or $-$.

WE DO

E1 $-5 + 2 \times 3$

YOU DO

1 $-1 + 2 \times 5$

$= -1 + 10$

$= 9$

2 $-10 + 3 \times 2$

$= -10 + 6$

$= \underline{\quad}$

3 $6 - 4 \times 3$

$= \underline{\quad} - \underline{\quad}$

$= \underline{\quad}$

4 $-8 + 12 \div 4$

$= \underline{\quad} + \underline{\quad}$

$= \underline{\quad}$

5 $-7 - 10 \div 2$

$= \underline{\quad} - \underline{\quad}$

$= \underline{\quad}$

6 $4 \times (-3) + 5$

$= \underline{\quad} + \underline{\quad}$

$= \underline{\quad}$

Step 2 Brackets first

HOW TO EASY

Work out the brackets first.
 $(-3 + 7) \times 2 = 4 \times 2 = 8$

Brackets first. Then the rest.

WE DO

E2 $(-2 + 5) \times 3$

YOU DO

7 $(-4 + 6) \times 5$

$= 2 \times 5$

$= 10$

8 $(1 - 7) \times 2$

$= -6 \times 2$

$= \underline{\quad}$

9 $(-3 - 2) \times 4$

$= \underline{\quad} \times \underline{\quad}$

$= \underline{\quad}$

10 $(-8 + 2) \div 3$

$= \underline{\quad} \div \underline{\quad}$

$= \underline{\quad}$

11 $-2 \times (5 - 9)$

$= \underline{\quad} \times \underline{\quad}$

$= \underline{\quad}$

12 $(-9 - 6) \div (-5)$

$= \underline{\quad} \div \underline{\quad}$

$= \underline{\quad}$

Step 3 $\times$ and $\div$ before $+$ and $-$

HOW TO MEDIUM

Do every $\times$ and $\div$ first. Then $+$ and $-$, left to right.
 $-10 \div 2 + 3 \times (-4) = -5 + (-12) = -17$

Do both $\times$ and $\div$ in the first line.

WE DO

E3 $2 \times (-3) + 12 \div 4$

YOU DO

13 $3 \times (-2) + 8 \div 2$

$= -6 + 4$

$= -2$

14 $-12 \div 3 - 2 \times 5$

$= -4 - 10$

$= \underline{\quad}$

15 $4 \times (-5) + (-18) \div 6$

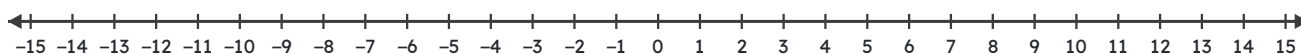
$= \underline{\quad} + (\underline{\quad})$

$= \underline{\quad}$

16 $-3 \times (-4) - 20 \div (-5)$

$= \underline{\quad} - \underline{\quad}$

$= \underline{\quad}$



Step 4 Left to right

HOW TO MEDIUM

Same rank? Work from **left to right**.

$$-12 \div 3 \times 2 = -4 \times 2 = -8$$

$$5 - 8 + 4 = -3 + 4 = 1$$

Work from left to right.

WE DO

E4 $-20 \div 5 \times 2$

E5 $3 - 7 + 2$

YOU DO

17 $-18 \div 6 \times 3$

$$= -3 \times 3$$

$$= -9$$

18 $6 - 10 + 3$

$$= -4 + 3$$

$$= \underline{\hspace{2cm}}$$

19 $24 \div (-4) \times 2$

$$= \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

20 $-2 - 5 + 9$

21 $-30 \div (-5) \times (-2)$

Step 5 Putting it together

HOW TO CHALLENGING

One step a line. Copy down everything you have not worked out yet.

$$-2 - (3 + (-5)) \times 4 = -2 - (-2) \times 4$$

$$= -2 - (-8) = 6$$

Brackets, then $\times$ and $\div$, then $+$ and $-$.

WE DO

E6 $5 - (2 - 6) \times 3$

YOU DO

22 $-4 + (1 - 3) \times 5$

$$= -4 + (-2) \times 5$$

$$= -4 + (-10)$$

$$= -14$$

23 $10 - (-2 + 6) \times 3$

$$= 10 - 4 \times 3$$

$$= 10 - 12$$

$$= \underline{\hspace{2cm}}$$

24 $(-8 + 2) \div 3 - 4$

$$= \underline{\hspace{2cm}} \div \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

25 $-3 \times (4 - 7) + (-10)$

Step 6 Is it right?

HOW TO CHALLENGING

Work it out yourself first.

Then compare. Say yes or no, and give the right answer.

Kim says $-3 + 4 \times 2 = 5$. 4×2 first: $-3 + 8 = 5$. Yes, Kim is right.

Is it right? Explain your answer.

WE DO

E7 Tom says $-2 + 3 \times 4 = 4$. Is he right?

YOU DO

26 Mia says $6 - 2 \times (-3) = 12$. Is she right?

$$2 \times (-3) = -6 \text{ first}$$

$$6 - (-6) = 12$$

Yes. Mia is right.

27 Ben says $-12 \div 2 \times 3 = -2$. Is he right?

$$\text{left to right: } -12 \div 2 = -6$$

$$-6 \times 3 = -18$$

Yes / No. The answer is ____.

28 Ali says $(-5 + 3) \times 2 = -4$. Is he right?

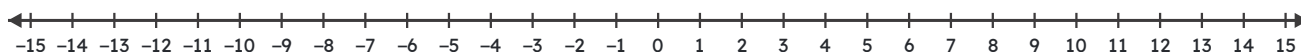
$$\text{brackets first: } \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

Yes / No. _____

29 Jo says $-4 + 6 \times 3 = 6$. Is Jo right?

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Step 1 The (-) key

HOW TO EASY

Use (-) for a negative number.

Use - to subtract.

$-35 + 12 = -23$. Press:

(-) 3 5 + 1 2 =

Write the keys you press.
Then the answer.

WE DO

E1 $-48 + 25$

YOU DO

1 $-63 + 41$

2 $-27 - 38$

(-) $63 + 41 =$

(-) $27 - 38 =$

$= -22$

$=$

3 $-150 + 86$

4 $45 - 112$

(-) $150 + 86 =$

$=$

5 $-208 - 97$

6 $-76 + 176$

Step 2 $\times$ and $\div$ with negatives

HOW TO EASY

Put brackets around a negative that comes after $\times$ or $\div$.

$-24 \times (-15) = 360$. Press:

(-) 2 4 $\times$ ((-) 1 5) =

Write the keys you press.
Then the answer.

WE DO

E2 -18×12

YOU DO

7 -23×15

(-) $23 \times 15 =$

$= -345$

8 $36 \times (-14)$

$36 \times (-14) =$

$=$

9 $-540 \div 12$

(-) $540 \div 12 =$

$=$

10 $-27 \times (-31)$

11 $-1\,196 \div (-26)$

Step 3 Check the sign first

HOW TO MEDIUM

Before you press =, use the sign rules: + or -?

If the calculator disagrees, check your keys.

-48×25 : different signs: -. The answer is $-1\,200$.

Choose the sign first.
Then use the calculator.

WE DO

E3 -34×29

YOU DO

12 $-52 \times (-18)$

same signs: +

$= 936$

13 $65 \times (-24)$

different signs: -

$=$

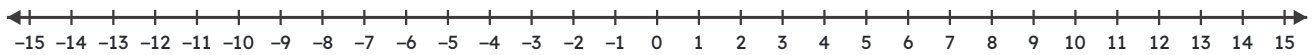
14 $-3\,024 \div 56$

different signs: -

$=$

15 $-2\,491 \div (-47)$

16 -125×16



Step 4 Brackets on the calculator

HOW TO MEDIUM

Press (and) for brackets. The calculator follows the order of operations.

$$(-35 - 47) \times 12 = -984$$

(() 3 5 - 4 7) × 1 2 =

Write the value of the brackets. Then the answer.

WE DO

$$E4 \quad (-28 + 13) \times 6$$

YOU DO

$$17 \quad (-45 - 19) \times 5$$

bracket: -64

= -320

$$18 \quad (36 - 81) \times (-4)$$

bracket: -45

= ____

$$19 \quad (-120 + 48) \div 8$$

bracket: ____

= ____

$$20 \quad (-67 - 58) \div (-25)$$

Step 5 Estimate, then calculate

HOW TO CHALLENGING

Round to the nearest 10 and work it out in your head. Then check on the calculator.

$$-48 \times 21 \approx -50 \times 20 = -1\,000.$$

Calculator: -1 008.

Estimate first. Then use the calculator.

WE DO

$$E5 \quad -39 \times 21$$

YOU DO

$$21 \quad -52 \times 19$$

$$\approx -50 \times 20 = -1\,000$$

= -988

$$22 \quad -61 \times (-29)$$

$$\approx -60 \times (-30) = 1\,800$$

= ____

$$23 \quad 78 \times (-41)$$

$$\approx ___ \times (___) = ___$$

= ____

$$24 \quad -403 + 198$$

Step 6 Calculator problems

HOW TO CHALLENGING

Write a number sentence first. Then use the calculator.

\$250, then 6 payments of \$45:
 $250 - 6 \times 45 = -20$. The balance is **-\$20**.

Write a number sentence. Then use the calculator.

WE DO

E6 An account has \$120. Four payments of \$38 come out. What is the balance?

YOU DO

25 A submarine at -185 m rises 12 m a minute for 9 minutes. Where is it now?

$$-185 + 9 \times 12$$

= -77 m

26 It is -14°C. It warms 3°C an hour for 7 hours. What is the temperature?

$$-14 + 7 \times 3$$

= ____°C

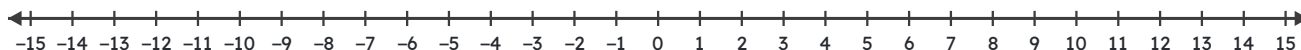
27 An account has \$85. Seven payments of \$26 come out. What is the balance?

$$___ - ___ \times ___$$

= ____

28 A diver at -42 m goes down 6 m a minute for 8 minutes. Where is she now?

NAME _____ CLASS _____ DATE _____

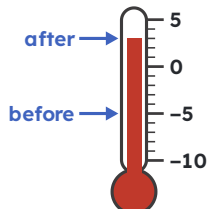


Step 1 Temperature

HOW TO EASY

A rise is +. A fall is -.

-5°C, then a rise of 8°C: $-5 + 8 = 3^{\circ}\text{C}$.



Write a sum. Then the new temperature.

WE DO

E1 -6°C, rises 4°C

YOU DO

1 -3°C, rises 9°C

2 2°C, falls 5°C

$-3 + 9$

$2 - 5$

$= 6^{\circ}\text{C}$

$= ___\text{ }^{\circ}\text{C}$

3 -1°C, falls 6°C

4 -8°C, rises 8°C

$______ - ______$

$= ___\text{ }^{\circ}\text{C}$

5 -10°C, falls 4°C

6 -12°C, rises 15°C

$______ - ______$

$______ - ______$

Step 2 Above and below sea level

HOW TO EASY

Above sea level is +. Below is -.

Distance apart = higher - lower.

A gull at 20 m and a diver at -10 m: $20 - (-10) = 30\text{ m}$.

How far apart are they?
Write a subtraction.

WE DO

E2 A gull at 20 m and a diver at -5 m

YOU DO

7 A boat at 0 m and a diver at -12 m

$0 - (-12)$

$= 12\text{ m}$

8 A cliff top at 30 m and a diver at -8 m

$30 - (-8)$

$= ___\text{ m}$

9 A plane at 150 m and a submarine at -60 m

$______ - (______)$

$= ___\text{ m}$

10 A diver at -5 m and a wreck at -22 m

11 A bird at 42 m and a shark at -37 m

$______ - ______$

$______ - ______$

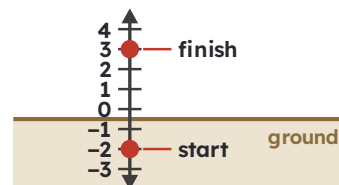
Step 3 Lifts and floors

HOW TO MEDIUM

The ground floor is level 0.

Basements are negative.

Level -2, up 5 floors: $-2 + 5 = \text{level } 3$.



Write a sum. Then the level.

WE DO

E3 A lift at level -1 goes up 4 floors. Which level is it on?

YOU DO

12 A lift at level 2 goes down 5 floors. Which level is it on?

$2 - 5$

$= \text{level } -3$

13 A lift at level -3 goes up 8 floors. Which level is it on?

$-3 + 8$

$= \text{level } ______$

14 A lift at level 5 goes down 9 floors. Which level is it on?

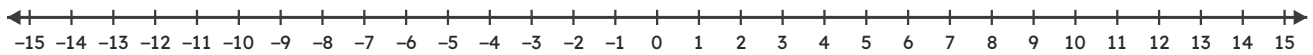
$______ - ______$

$= \text{level } ______$

15 A lift at level -2 goes down 2, then up 7. Which level is it on?

$______ - ______$

$______ - ______$



Step 4 Money

HOW TO **MEDIUM**
Money in is +. Money out is -.
Below \$0 means you owe money.
\$20, then pay \$35: $20 - 35 = -15$.
The balance is -\$15.

Write a sum. Then the new balance.

WE DO

E4 The balance is \$30. You pay \$45.

YOU DO

16 The balance is \$12. You pay \$25.

12 - 25

= -\$13

17 The balance is -\$20. You pay in \$50.

-20 + 50

= ____

18 The balance is -\$15. You pay \$30.

____ - ____

= ____

19 The balance is \$40. You pay \$25, then \$30.

20 The balance is -\$60. You pay in \$35, then \$40.

Step 5 Several steps

HOW TO **CHALLENGING**
Break it into steps. One number sentence a step.
-40 m, up 3 m a minute for 5 minutes: $3 \times 5 = 15$, $-40 + 15 = -25$ m.

Write a number sentence for each step.

WE DO

E5 It is -6°C . It warms 2°C an hour for 5 hours. How warm is it now?

YOU DO

21 A diver at -30 m rises 4 m a minute for 6 minutes. Where is she?

$4 \times 6 = 24$, $-30 + 24$

= -6 m

22 It is 5°C . It cools 3°C an hour for 4 hours. How cold is it now?

$4 \times (-3) = -12$, $5 + (-12)$

= ____ $^{\circ}\text{C}$

23 A win scores +3, a loss -2. A team wins 4 and loses 7. What is its score?

____ + ____ $\times$ (____) = ____ + (____)

= ____

24 Sam has \$50 and pays \$12 a week for 6 weeks. What is the balance?

Step 6 Is it right?

HOW TO **CHALLENGING**
Work it out yourself first.
Then compare. Say yes or no, and give the right answer.
Kim says -4°C , up 6°C , is 2°C . $-4 + 6 = 2$. Yes.

Is it right? Explain your answer.

WE DO

E6 Tom says level -3, up 5 floors, is level -8. Is he right?

YOU DO

25 Mia says 12 m and -9 m are 3 m apart. Is she right?

12 - (-9)

= 21

No. They are 21 m apart.

26 Ben says -\$20, then \$35 in, leaves \$15. Is he right?

-20 + 35

= 15

Yes / No. _____

27 Ali says 3°C , falling 8°C , gives -5°C . Is he right?

____ - ____

= ____

Yes / No. _____

28 Jo says a score of -4 is better than -9. Is Jo right?